Chapter 5:Advanced SQL

Programmable SQL: Function, Procedure and Trigger

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Chapter Outline

PL/SQL Block and conditional statement

Conditional Construct, Iteration

PL/SQL Block

Block: Smallest unit of code

In PL/SQL, as in most other procedural languages, **the smallest meaningful grouping of code** is known as a block. A block is a unit of code that provides **execution and scoping boundaries for variable declarations and exception handling**.

Blocks: Anonymous

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It has no Name

It cannot be called by any other blockit doesnt have a handle for reference. Instead, anonymous blocks serve as containers that execute PL/SQL statements. It is **not stored** in the database and **can not be reused**. (one-time execution).

```
BEGIN

DBMS_OUTPUT.put_line ('Hello World!');

END;
```

Block: Named

It has Name

While anonymous PL/SQL blocks are indispensable, the majority of code you write will be in named blocks. **Examples:** Functions and Procedure. It is stored in the database and can be reused.

Block: Equivalence in Oracle and Postgre SQL

In Oracle, you start an anonymous block with **DECLARE** or **BEGIN**. In PostgreSQL, you must use the **DO** keyword. By default, **DO** executes code in the plpgsql language.

Oracle

```
DECLARE
v_count NUMBER;
BEGIN
SELECT count(*) INTO
    v_count FROM
    transactions;
DBMS_OUTPUT.PUT_LINE('Total:
    ' || v_count);
END;
```

Postgre SQL

```
DO $$
DECLARE
v_count INTEGER;
BEGIN
SELECT count(*) INTO
    v_count FROM
    transactions;
RAISE NOTICE 'Total: %',
    v_count;
END $$;
```

Conditional Statement: Equivalence in Oracle and Postgre SQL

Oracle

```
DECLARE
v_balance NUMBER := 150;
BEGIN
  IF v_balance > 100 THEN
  DBMS_OUTPUT.PUT_LINE('High Balance');
  ELSIF v_balance > 50 THEN
  DBMS_OUTPUT.PUT_LINE('Medium Balance');
  ELSE
  DBMS_OUTPUT.PUT_LINE('Low Balance');
  END IF;
END;
```

Postgre SQL

```
DO $$
DECLARE
v_balance NUMERIC := 150;
BEGIN
  IF v_balance > 100 THEN
  RAISE NOTICE 'High Balance';
  ELSIF v_balance > 50 THEN
  RAISE NOTICE 'Medium Balance';
  ELSE
  RAISE NOTICE 'Low Balance';
  END IF;
END $$;
```

Iteration: Equivalence in Oracle and Postgre SQL

Oracle

```
BEGIN
FOR i IN 1..5 LOOP
DBMS_OUTPUT.PUT_LINE('Iteration:
    ' || i);
END LOOP;
END;
```

Postgre SQL

```
DO $$
BEGIN
FOR i IN 1..5 LOOP
RAISE NOTICE
    'Iteration: %', i;
END LOOP;
END $$;
```

Function

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Functions (Cont.): Equivalence in Oracle and Postgre SQL

Oracle

```
CREATE OR REPLACE FUNCTION
    calculate_tax(p_amount
    NUMBER)
RETURN NUMBER
IS
BEGIN
RETURN p_amount * 0.10;
END;
/
```

Postgre SQL

```
CREATE OR REPLACE FUNCTION
    calculate_tax(p_amount
    NUMERIC)
RETURNS NUMERIC
AS $$
BEGIN
RETURN p_amount * 0.10;
END;
$$ LANGUAGE plpgsql;
```

A few points about Functions

- IN parameter (like conventional parameter), OUT parameter (not used here. used for writing the result back to the caller)
- No datatype precision is allowed in parameter and return type (**...return varchar2(20)** is not valid) **applicable both in Oracle and Postgre**
- Function Name and parameters should be **self-explanatory**. (do not use abc as function name, x, y , z as parameters)
- Remember: in Oracle **Type Strictness** is relaxed while in Postgre it is very strict often needs manual casting.

Function: A simple example

```
CREATE OR REPLACE FUNCTION get_total_sales
RETURN NUMBER
IS
v_total_sales NUMBER := 0;
BEGIN
-- get total sales
SELECT SUM(unit_price * quantity)
INTO l_total_sales
FROM order_items
INNER JOIN orders USING (order_id)
WHERE status = 'Shipped';


-- return the total sales
RETURN v_total_sales;
END;
```

Calling a function

There are 3 general options:

1. It can be readily tested by a fixed parameter from dual:
   ```
   select get_total('CSE') from dual;
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   DBMS_OUTPUT.PUT_LINE('Sales: ' || get_total_sales);
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2. You can test it by passing it as a parameter in put_line function:
   ```
   DBMS_OUTPUT.PUT_LINE('Sales: ' || get_total_sales);
   ```
3. It can be selected from a table values, the parameter values can be passed dynamically in the select statement.
   ```
   select id, get_cgpa(sid) from students where dept='CSE';
   ```

DECODE function

```
DECODE(expr, search, result
[, search, result ]...
[, default ]
)
```

DECODE()

DECODE compares expr to each search value one by one. If expr is equal to a search, then Oracle Database returns the corresponding result. If no match is found, then Oracle returns default. If default is omitted, then Oracle returns null.

DECODE by example

```
SELECT product_id,
DECODE (warehouse_id, 1, 'Southlake',
2, 'San Francisco',
3, 'New Jersey',
4, 'Seattle',
'Non domestic')
"Location of inventory" FROM inventories
WHERE product_id < 100;
```

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- The RANK() function is useful for **top-N** and **bottom-N** queries.

RANK() Syntax and Example:

```
RANK()
OVER ([ query_partition_clause ] order_by_clause)

Example:

select sid,name,cgpa,
rank() over (order by cgpa desc) position
from students;
```

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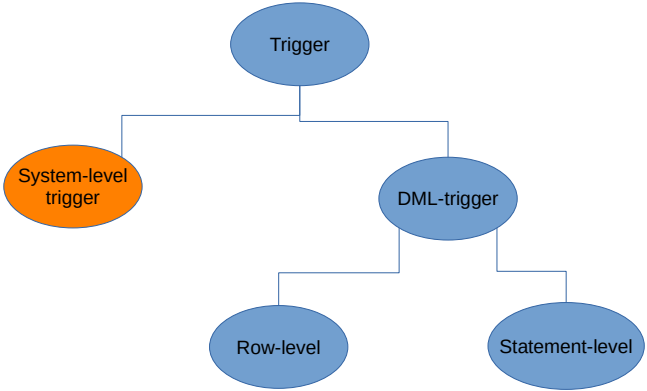
- To design a trigger mechanism, we must:
  - Specify the conditions under which the trigger is to be executed.
  - Specify the actions to be taken when the trigger executes.
- Since trigger is event-oriented, it is not explicitly called like functions rather it automatically fires wherever the condition is met.

Trigger : Broad Classification



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So, they have one in common: **it is executed for each possible record's change.**

Row-level Trigger (Cont.), Syntax

Indicator: **for each row** clause

```
CREATE OR REPLACE TRIGGER trigger_name
BEFORE | AFTER
INSERT OR DELETE OR UPDATE OF column1, column2,
ON table_name
FOR EACH ROW
REFERENCING OLD AS old_name
NEW AS new_name
WHEN (condition)
DECLARE

BEGIN

EXCEPTION

END;
```

Statement-level Trigger

It works exactly in the same way as except it is executed only once no matter how many records are affected by the DML statement.

Syntax: Just omit the **for each row** option.